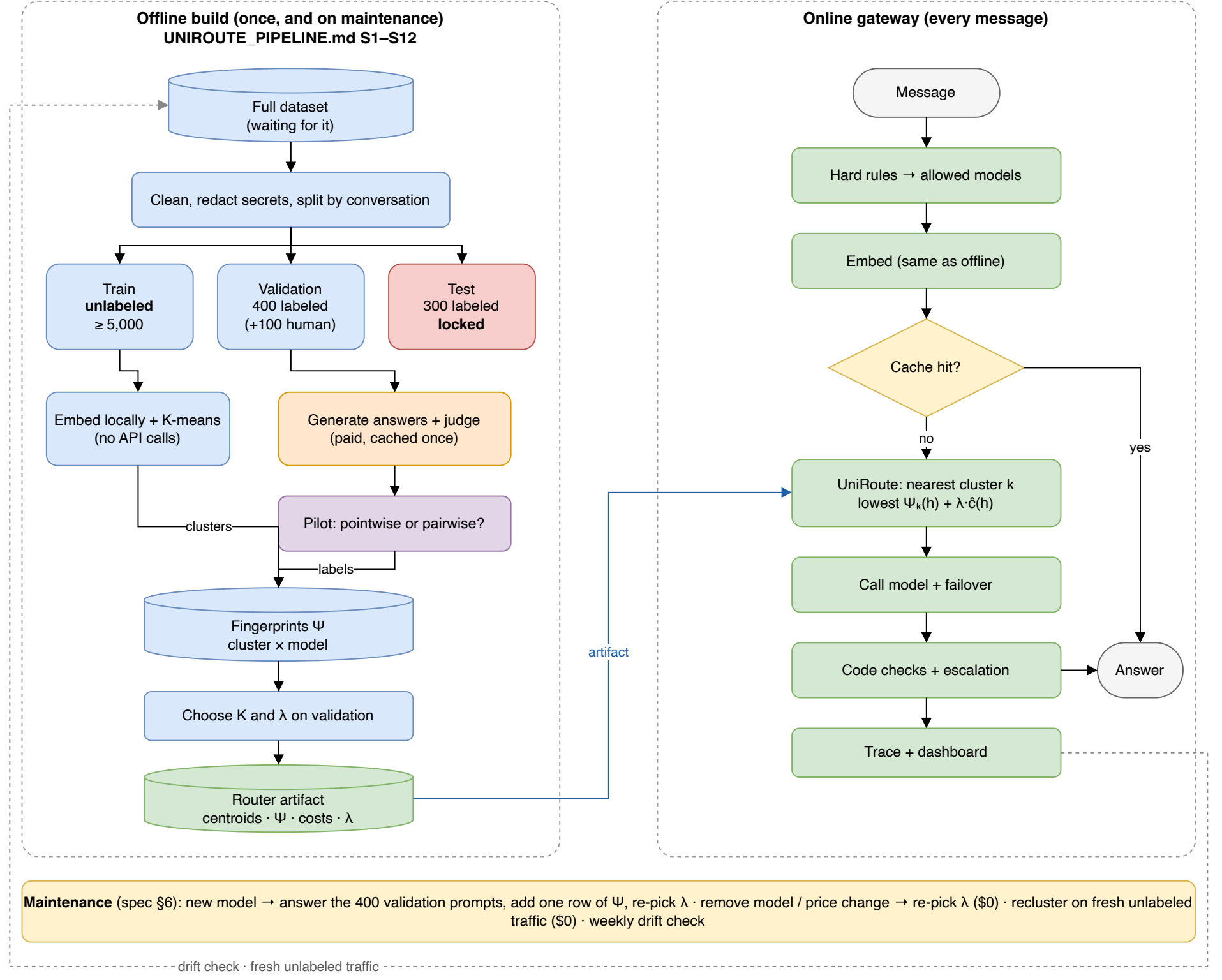

Offline build (once, and on maintenance)
UNIROUTE_PIPELINE.md S1–S12
Full dataset
(waiting for it)
Clean, redact secrets, split by conversation
Train
unlabeled
≥ 5,000
Validation
400 labeled
(+100 human)
Test
300 labeled
locked
Embed locally + K-means
(no API calls)
Generate answers + judge
(paid, cached once)
clusters
Pilot: pointwise or pairwise?
labels
Fingerprints Ψ
cluster × model
Choose K and λ on validation
Router artifact
centroids · Ψ · costs · λ
artifact
Online gateway (every message)
Message
Hard rules → allowed models
Embed (same as offline)
Cache hit?
no
yes
UniRoute: nearest cluster k
lowest Ψk(h) + λ·ĉ(h)
Call model + failover
Code checks + escalation
Answer
Trace + dashboard
Maintenance (spec §6): new model → answer the 400 validation prompts, add one row of Ψ, re-pick λ · remove model / price change → re-pick λ ($0) · recluster on fresh unlabeled traffic ($0) · weekly drift check
drift check · fresh unlabeled traffic